Hyun Joon Cha

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Research Interest | *Reinforcement Learning, Data Infrastructure for AI, Digital Twin, AI for Science*

Education

Seoul National University Mar 2019 – Present

- MS. Mechanical Engineering

Korea University Mar 2019 – Feb 2025

- B.S. Chemical and Biological Engineering

Research Experiences

Renewable Energy Conversion Lab, Seoul National University *Jun 2024 – Present*
Graduate Student. Supervisor: Prof. Suk Won Cha

- Designed Reinforcement Learning-based fuel cell power controller for Fuel Cell Hybrid Electric Vehicles to maximize fuel economy while minimizing fuel cell and battery degradation that led to

System Bioengineering Lab, Korea University *Nov 2023 – May 2024*
Undergraduate researcher. Supervisor: Prof. Min Kyu Oh

- Engineered a low-cytotoxicity Cas9-BD system for high-GC Streptomyces, reducing off-target effects and enabling multiplexed BGC refactoring that led to a 2.4-fold increase in specialized metabolite production.

Publications and Conferences

1. Hyun Joon Cha, Donghwan Sung, Hyunjun Yang, Gwangeon Lee, Suk Won Cha*, "Reinforcement Learning-Based Energy Management System for Fuel Cell Hybrid Electric Vehicles Considering Fuel Cell Degradation", 2025 Korean Society of Automotive Engineers (KSAE) Autumn Conference [1st author]

Selected Honors and Awards

Chemical Engineering Capstone Design Top Award, *Korea University* *Dec 2024*

- Optimized sensor type and deployment strategies in underground garages to detect immediate thermal runaway

Selected Projects

Global Human Resources Program for Core Technology of Future Mobility, *KIAT* *May 2026 – Oct 2026*

- 6-month scholarship for visiting research

In-line Defect Battery Detection using EIS, *Mona* *Jun 2025 – Nov 2025*

- Designed a defect battery detection algorithm by analyzing large-scale electrochemical parameter data from battery cell production lines

Extracurricular Activities and Leadership

TAKSA, Table Tennis Club, Korea University *Mar 2019 – Feb 2025*

- President of Table Tennis Club; participated in numerous competitive tournaments

Skills Summary

Programming Languages: Python, C++, MATLAB, PostgreSQL

Frameworks/Tools: Docker, PyTorch, Tensorflow, Simulink, WSL2

Languages: Korean (Native), English (Fluent, iBT TOEFL 5.5, 110)